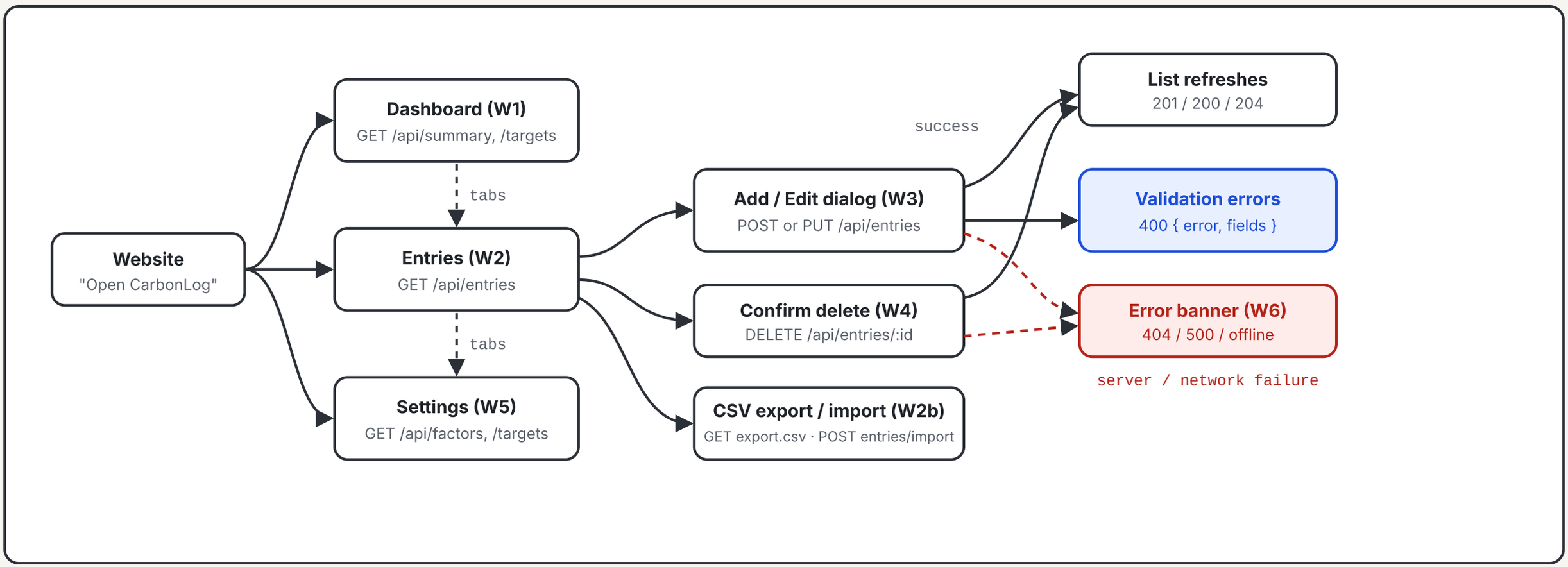
CarbonLog – annotated wireframes

Low-fidelity screens of the CarbonLog app as it is implemented in `public/carbonlog.html`. Numbers mark controls and states; each note names the requirement from `docs/business-case.md` (R1–R14) and the API call behind it.

1 numbered callout R4 requirement it serves red error state

W0 Navigation and flow

One page with three tabs. Adding and editing happen in a dialog, so the user never loses the list. Every action calls the REST API and then refreshes what is on screen.



W1 Dashboard

The first thing the owner-manager sees: the total, the biggest source and the trend, for any date range.

1 Tabs
Dashboard, Entries, Settings. The active tab is filled; one page, no reload. "←

← Website **CarbonLog** ● Live · updated 14:32:05 (7) Dashboard | Entries | Settings (1)

Dashboard

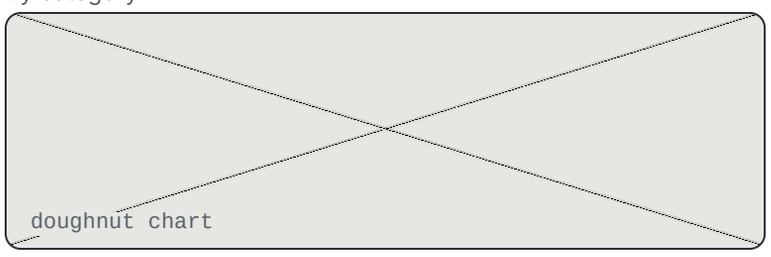
Know your footprint in minutes: log activities, see where your emissions come from.

From 01.01.2026 To 31.03.2026 **Apply** **Clear** (2)

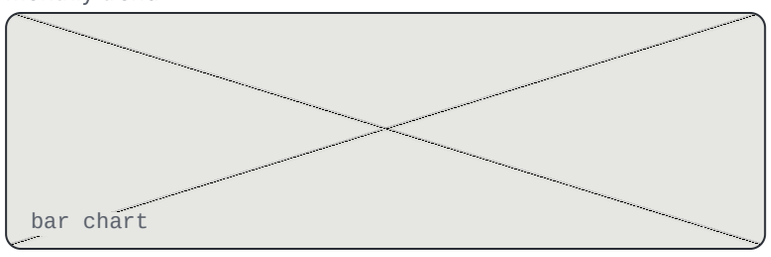
Total emissions	Entries	Largest source (3)
1,284 kg CO_2	**17**	**Electricity**

(4)

By category



Monthly trend



(5)

Reduction target

Target: reduce by 10% versus 2025 (2,100 kg). So far this year: 1,284 kg, which is 39% below the baseline.

(6)

CarbonLog provides **estimates**, not certified emission reports. Emission factors are approximate...

Website" returns to the marketing site.

(2) Date range
From/To plus Apply; Clear shows all time. Sends `GET /api/summary?from=&to=`. R7

(3) Key figures
Total, number of entries, largest source. Numbers come from the server, never computed in the browser. R7

(4) Charts
Share by category and a monthly trend (Chart.js). If the charts cannot load, the numbers above still show. R7 R12

(5) Target progress
Compared with the previous calendar year; the bar turns orange when the year is above target. `GET /api/targets/:year` R8

(7) Live indicator
A green dot with the time of the last change. The page checks the server every 5 seconds in the background and refreshes the figures only when something changed. "Offline · retrying" turns it red. `GET /api/changes` R14

(6) Estimate disclaimer
Always visible, so nobody mistakes the figures for a certified report. Scope

W1b Dashboard states: loading, empty, charts unavailable

The page never stays blank: each state tells the user what is happening and what to do next.

localhost:3000/carbonlog.html

(1)

Loading...

(1) Loading
Shown while `/api/summary` is on its way. R12

(2) Empty

No entries in this period yet. Add your first activity on the Entries tab.

Charts could not be loaded (Chart.js CDN unreachable). Numbers are still shown above.

Reduction target
No target set for this year. You can set one under Settings.

No entries in the chosen range; points the user to the next step. R12

3 Charts unavailable
The chart library comes from a CDN; offline the app degrades instead of breaking. R12

4 No target yet
Explains where to set one. If the previous year has no entries, it says progress cannot be calculated. R8

W2 Entries

The list of everything logged, newest first, with edit and delete on each row.

localhost:3000/carbonlog.html

← Website **CarbonLog** ● Live · updated 14:32:05 Dashboard Entries Settings

Entries

Category
All categories ▾ Add entry Export CSV Import CSV Download template

Date	Category	Amount	CO_2 (kg)	Note	
2026-03-15	Electricity	100 kWh	38	Office, March bill	Edit Delete
2026-03-08	Train travel	420 km	14.7	Trade fair Munich	Edit Delete
2026-02-28	Natural gas	1,200 kWh	241.2	Heating	Edit Delete

No entries yet. Click "Add entry" to log your first activity.

1 Add entry
Opens the dialog (W3). R1

2 Export CSV
Downloads every entry: `GET /api/export.csv`. Formulas in notes are neutralised. R10

3 Table
CO_2 is the value stored when the entry was saved; changing a factor later does not change it. Edit opens W3 pre-filled; Delete opens W4. R5 R6 R9

4 Empty state
Shown instead of an empty table on first use. R12

5 Import CSV
Opens the file picker; the file text is sent to `POST /api/entries/import`. "Download template" gives a ready-made file with the right columns. Result: see W2b. R13

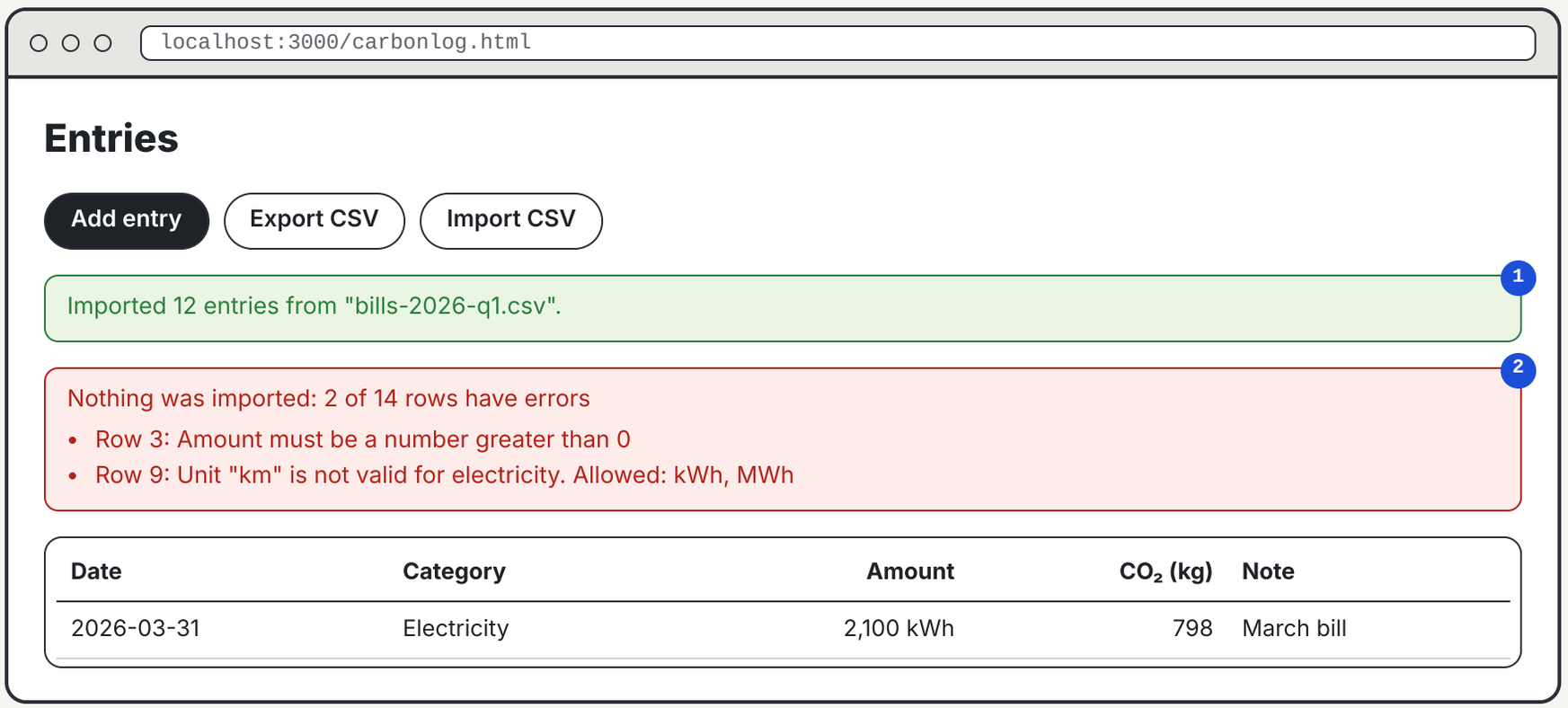
6 Live indicator

Entries added or changed on another device or tab appear here within about 5 seconds, with no reload and never while the dialog is open. R14

7 **Category filter**
"All categories" plus the six categories. Choosing one sends `GET /api/entries?category=…`; the filter stays on during the live refresh. If nothing matches: "No Train travel entries yet. Choose "All categories" to see everything." Built on branch `feature/vignesh-category-filter` (pull request #1). R6

W2b CSV import: success, and nothing imported

The import is all-or-nothing. The user either sees how many entries were added, or a list of the exact rows to fix, and the data stays clean.

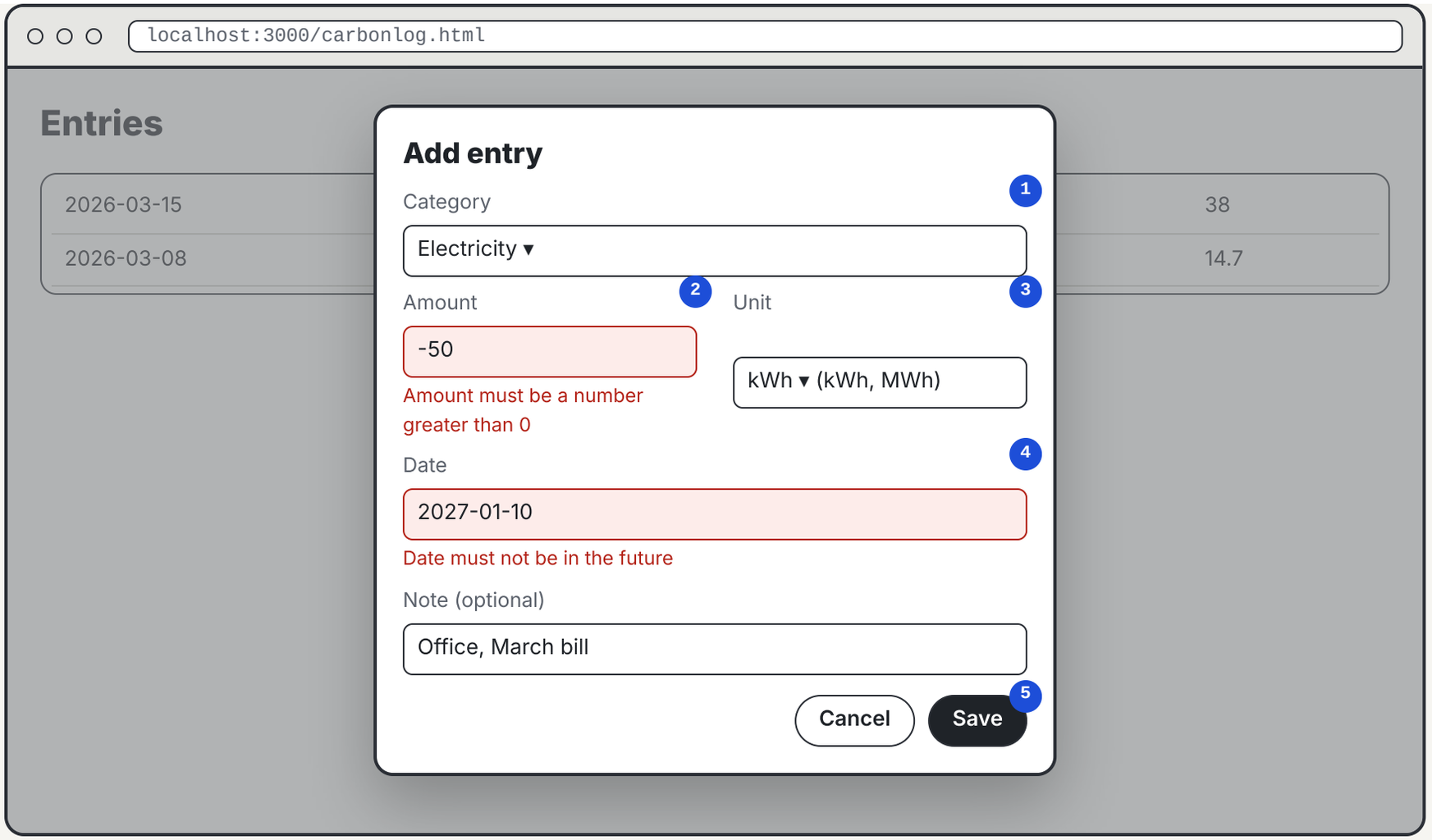


1 **Success**
`201 { imported: 12 }`; the list reloads with the new rows. CO_2 is calculated by the server for every row. R13

2 **Nothing imported**
`400` with each bad row: its row number in the file and the same messages as the form. Fix the file and import again. R4 R13

W3 Add / edit entry, with validation errors

The same dialog adds and edits. The server checks every field and answers with messages that appear under the right field.



1 **Category**
Six categories; choosing one limits the unit list to that category's units. R1

2 **Amount**
Must be a number > 0 and ≤ 1,000,000,000. Errors come back as `400 { error, fields }` and appear here. R4

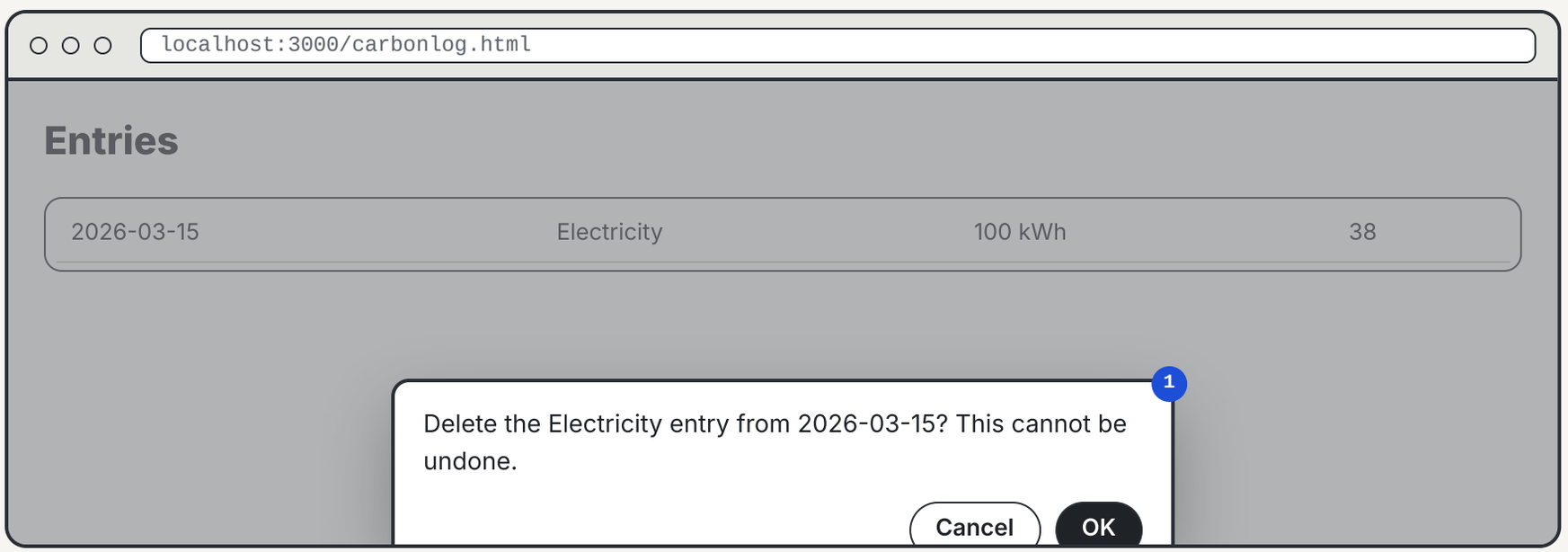
3 **Unit**
Only the category's units: kWh/MWh for energy, km/miles for travel, EUR for goods. Converted on the server. R2 R3

4 **Date**
A real calendar date, not in the future. R4

5 **Save**
New: `POST /api/entries` → 201. Edit: `PUT /api/entries/:id` recalculates CO_2. On success the dialog closes and the list refreshes. R1 R5

W4 Confirm delete

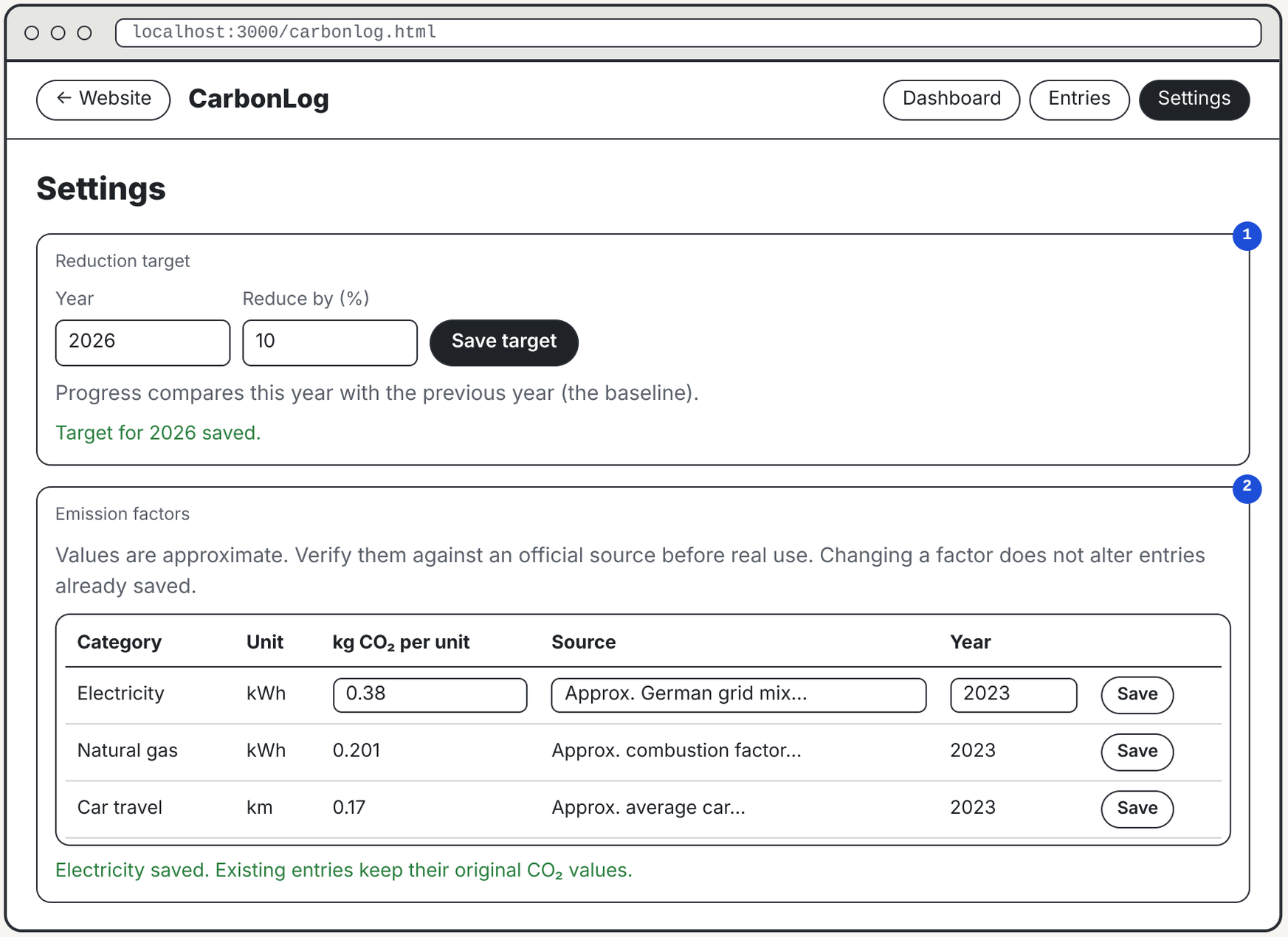
Deleting cannot be undone, so the user confirms first and sees exactly which entry goes.



1 **Browser confirm**
OK sends `DELETE /api/entries/:id` → 204, then the list and the dashboard refresh. Cancel changes nothing. R6

W5 Settings: reduction target and emission factors

Where the company sets its yearly goal and keeps the emission factors current, with a source and a year for each.

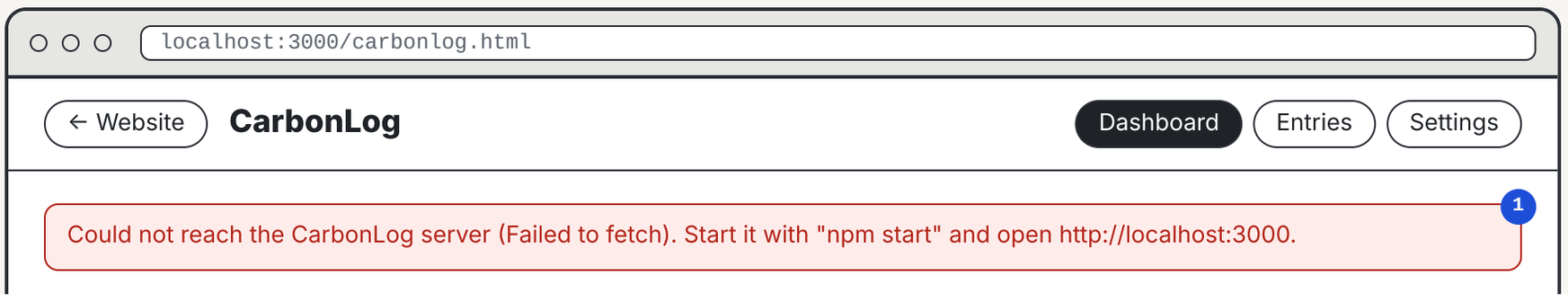


1 **Target**
Year 1990–2100 and 0–100 %. `PUT /api/targets/:year`; confirmation appears below the form. R8

2 **Factors**
Value, source and year are editable per row. `PUT /api/factors/:id`. The message makes clear that history is not rewritten. R9

W6 Server not reachable

If the Node.js server is not running, the user gets a clear message with the fix instead of a frozen page.



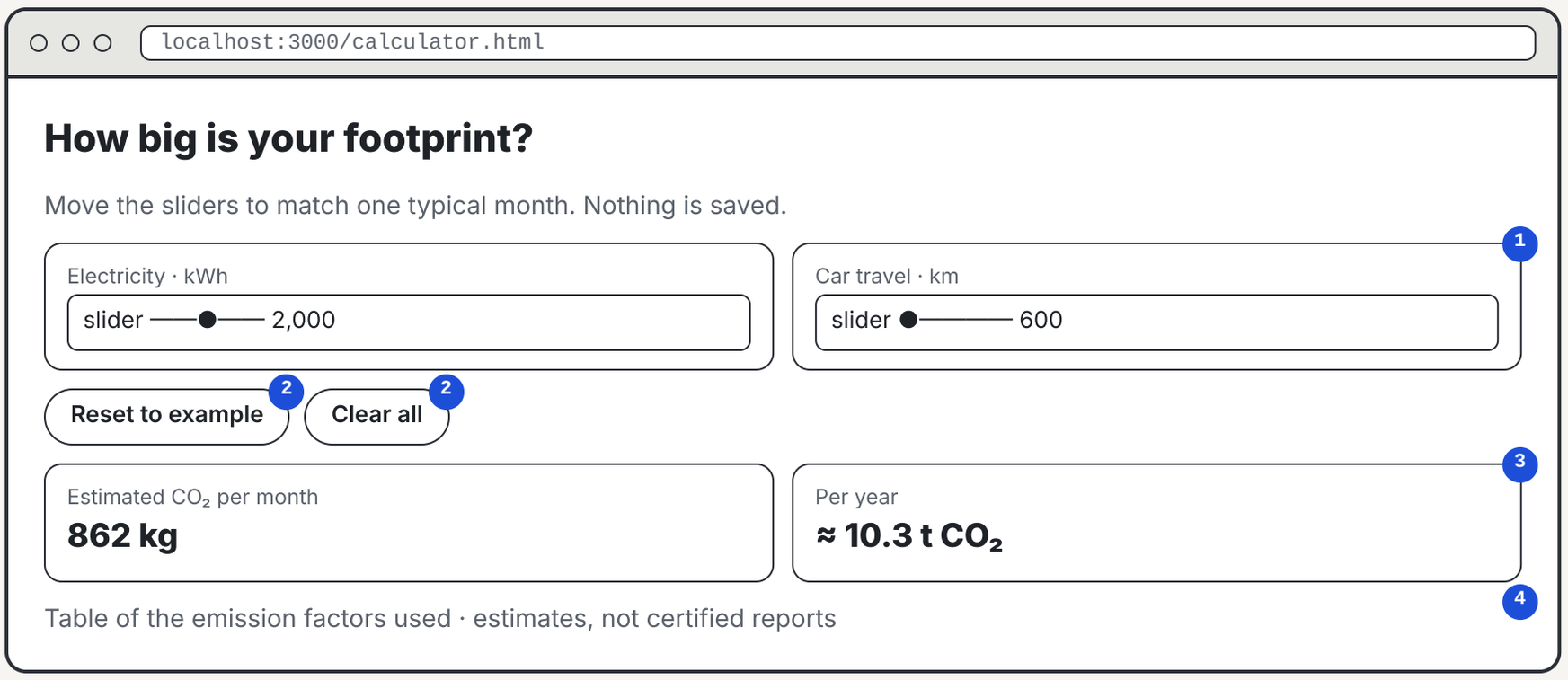
1 **Global error banner**
Used for network failures and server errors (404, 500). The `api()` helper turns every non-OK response into a readable message. R11 R12

Dashboard

–

W7 Website: Quick Check and Pricing

The marketing site around the app. The Quick Check lets a visitor estimate a footprint without logging in; Pricing explains the business model. Both are concept screens: no data is stored, and the plans are not enforced in the app (see scope decisions in the business case).

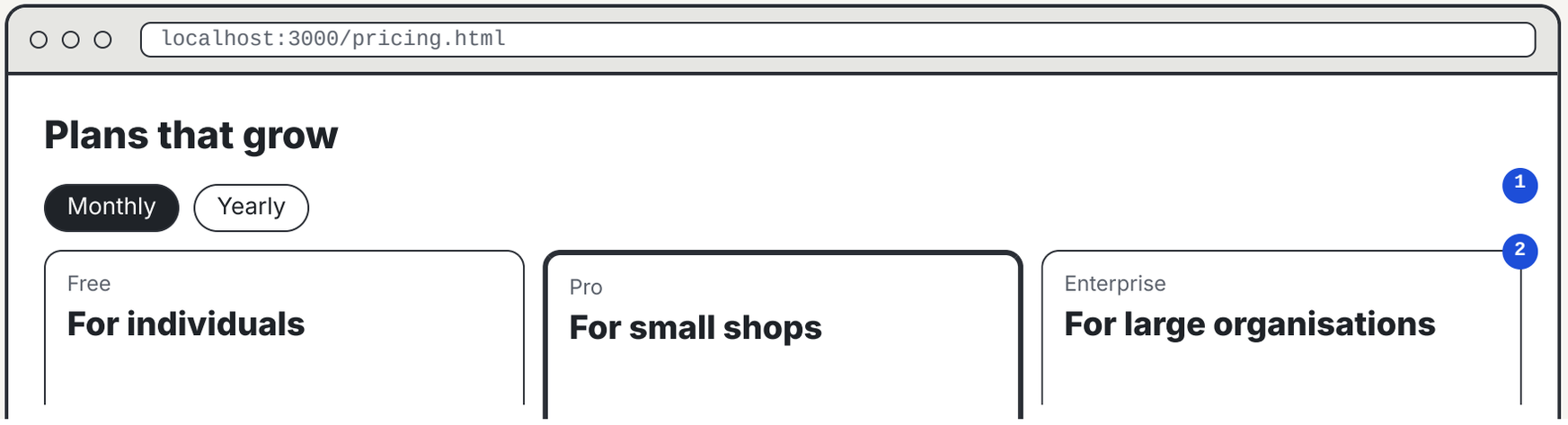


1 **Sliders**
One slider per emission source (electricity, natural gas, car, train, flights, purchased goods) for one typical month. The result updates as the visitor moves them; nothing is sent to the server.

2 **Reset / Clear**
Reset restores the example numbers; Clear empties every field.

3 **Result**
Estimated CO_2 per month and the yearly equivalent in tonnes, using the same factors as the app.

4 **Factors table**
The factors behind the estimate are shown, with the note that the figures are estimates.



1 **Billing cycle**
Monthly or yearly; the choice is carried to the checkout in the address (`?cycle=`).

2 **Three plans**
Free (individuals), Pro (small shops), Enterprise (large organisations). "Choose Pro" opens the checkout (W8).

Choose Pro

3

What each plan includes · Before you choose

3 **Details**
A comparison of what each plan includes and answers to common questions.

W8 Website: Pro checkout and success page

A demo of the purchase flow. It shows billing cycle, VAT and payment methods, but no payment is made, and no card data is stored or sent: the success page receives only the last four digits for the receipt.

localhost:3000/checkout.html?cycle=yearly

Upgrade to Pro

1

Billing cycle

Yearly ▾

Company and billing details

Work email

Company, address, country, VAT ID

2

Payment method

Card · IBAN

Card number · expiry · security code

Card number is not valid

3

Order summary

Net price + VAT for the chosen country = total

Pay

1 **Details**
Billing cycle and company data. The VAT rate follows the country, and a VAT ID changes the VAT line for business customers.

2 **Payment**
Card or bank (IBAN). Each field is checked in the browser (card number checksum, IBAN check); the message sits under the field.

3 **Pay**
Whichever method is chosen, the visitor lands on the success page.

localhost:3000/payment-success.html

Welcome to CarbonLog Pro

1

Receipt

Plan, billing cycle, total, payment method (last four digits only)

2

What happens next

Open CarbonLog

1 **Receipt**
Built from the order held in the browser session for this visit. No card data is passed on.

2 **Next steps**
A short list of what to do next, with a button into the app.